

EXPERT SYSTEMS

Expert systems (ES) is one of the prominent research domains of AI. It was introduced by the researchers at Stanford University, Computer Science Department. The expert systems are the computer applications developed to solve complex problems in a particular domain, at the level of extra-ordinary human intelligence and expertise.

Def:

ES is an intelligent computer program that uses knowledge and inference to solve a problem that are difficult enough to require a significant human expertise for their solution (prof. E O Nwachukwu)

An expert system is a computer program that represents and reasons with knowledge of some specialist subject with a view to solving problems or giving advice (Jackson, 1999)

Examples of expert systems

1. MYCIN

- Historically, the MYCIN system played a major role in stimulating research interest in rule-based expert systems.
- MYCIN is an expert system for diagnosing and recommending treatment of bacterial infections of the blood (such as meningitis and bacteremia). It was developed at Stanford University in California, USA, in the 1970s, and has become a template for many similar rule based systems. It is intended to support clinicians in the early diagnosis and treatment of meningitis, which can be fatal if not treated in time.

2. Prospector:

- PROSPECTOR is an expert system which was designed for decision making problems in mineral exploration. It uses a structure called an inference network to represent the data base. PROSPECTOR was written in 1978 by Richard O. Duda (1979).

Example of MYCIN

Here is an excerpt from a sample session for MYCIN:

1. What is the patient's name?
 - John Maina
2. Male or Female?
 - Male
3. Age?
 - 55
4. Have you obtained positive cultures indicating general type?
 - Yes.
5. What type of infection is it?
 - Primary bacteria.
6. When did symptoms first appear?
 - May 5, 2019
7. Let's call the most recent positive culture C1. From what site was C1 taken?
 - From the blood.
8. When?
 - May 9, 2019

9. Let's call the first significant organism from this culture U1. Do you know the identity of U1?
 - No.
10. Is U1 a rod or a coccus or something else?
 - Rod.
11. What in the gram stain of U1?
 - Gram-negative.
12. Have there been positive cultures from sites which are not associated with the infections about which you are seeking advice?
 - No.
13. Did U1 grow aerobically?
 - Yes.

When MYCIN is satisfied that it has a reasonably good grasp of the situation, MYCIN announces its diagnosis:

My recommendation will be based on the opinion that the identity of U1 may be:

1. Pseudomonas-aeruginosa
2. Klebsiella-pneumoniae.
3. E.coli
4. Bacteroides-fragilis.
5. Enterobacter.
6. Proteus-mirabilis.

After asking a few more questions about the patient, MYCIN announces its recommendation:

1. On a scale of 0 to 4, where the higher number indicates greater severity, how would you rate the patient's degree of sickness?
2. Does the patient have a clinically significant allergic reaction to any antimicrobial agent?
3. How much does the patient weigh?
4. My first recommendation is as follows: to cover for items 1, 2, 3, 5 and 6, give gentamycin using a dose of 119 mg (1.7 mg/kg) q8h IV [or IM] for 10 days. Modify dose in renal failure. Also, to cover for item 4, give clindamycin using a dose of 595 mg (8.5 mg/kg) q6h IV [or IM] for 14 days. If diarrhea or other gastrointestinal symptoms develop, check for pseudomembranous colitis.

It has the following organizational features:

i. Knowledge Representation:

- In the form of production rules implemented in LISP. K.R. is of the form – If condition, and and condition hold then draw conclusion, and and conclusion encoded in the data structure of LISP programming.

ii. Reasoning:

- Backward chaining, and forward reasoning uses certainty factors to reason with uncertain information. MYCIN uses backward chaining to discover what organisms were present. Then it uses forward chaining to reason from the organism to a treatment regime.

iii. Heuristics:

- When the general category of infection has been established, MYCIN examines each candidate diagnosis in a depth-first manner. Heuristics are used to limit the search,

including checking all premises of a possible rule to see if anyone of these is known to be false.

iv. Dialogue/Explanation:

- The dialogue is computer controlled, with MYCIN driving the consultation through asking questions. Explanations are generated by tracing back through the rules which have been fired. Both “how? ” and “why? ” explanation are supported.

Characteristics of Expert Systems

- High performance
- Understandable
- Reliable
- Highly responsive

Capabilities of Expert Systems

The expert systems are capable of:

- Advising
- Instructing and assisting human in decision making
- Demonstrating
- Deriving a solution
- Diagnosing
- Explaining
- Interpreting input
- Predicting results
- Justifying the conclusion
- Suggesting alternative options to a problem

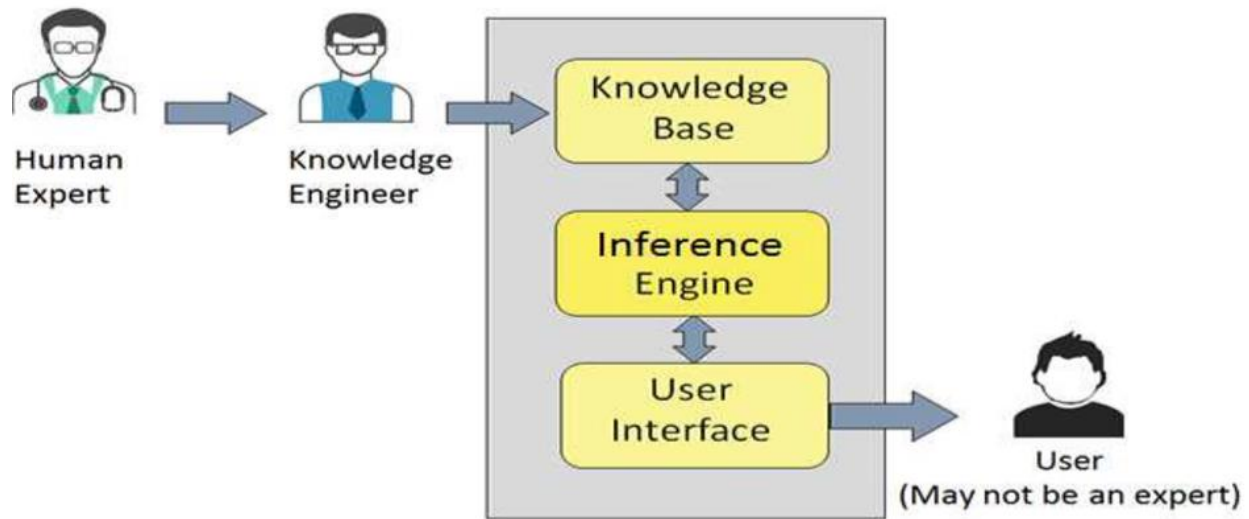
They are incapable of:

- Substituting human decision makers
- Possessing human capabilities
- Producing accurate output for inadequate knowledge base
- Refining their own knowledge

Components of Expert Systems

The components of ES include:

- Knowledge Base
- Inference Engine
- User Interface



1. Knowledge Base

It contains domain-specific and high-quality knowledge. Knowledge is required to exhibit intelligence. The success of any ES majorly depends upon the collection of highly accurate and precise knowledge.

What is Knowledge?

The data is collection of facts. The information is organized data and facts about the task domain. Data, information, and past experience combined together are termed as knowledge.

Components of Knowledge Base

The knowledge base of an ES is a store of both, factual and heuristic knowledge.

Factual Knowledge – It is the information widely accepted by the Knowledge Engineers and scholars in the task domain.

Heuristic Knowledge – It is about practice, accurate judgment, one's ability of evaluation, and guessing.

Knowledge representation

It is the method used to organize and formalize the knowledge in the knowledge base. It is in the form of IF-THEN-ELSE rules.

Knowledge Acquisition

The success of any expert system majorly depends on the quality, completeness, and accuracy of the information stored in the knowledge base. The knowledge base is formed by readings from various experts, scholars, and the Knowledge Engineers. The knowledge engineer is a person with the qualities of empathy, quick learning, and case analyzing skills.

He acquires information from subject expert by recording, interviewing, and observing him at work, etc. He then categorizes and organizes the information in a meaningful way, in the form of IF-THEN-ELSE rules, to be used by interference machine. The knowledge engineer also monitors the development of the ES.

2. Inference Engine

Use of efficient procedures and rules by the Inference Engine is essential in deducting a correct, flawless solution.

In case of knowledge-based ES, the Inference Engine acquires and manipulates the knowledge from the knowledge base to arrive at a particular solution.

In case of rule based ES, it:

- Applies rules repeatedly to the facts, which are obtained from earlier rule application.
- Adds new knowledge into the knowledge base if required.
- Resolves rules conflict when multiple rules are applicable to a particular case

To recommend a solution, the inference engine uses the following strategies:

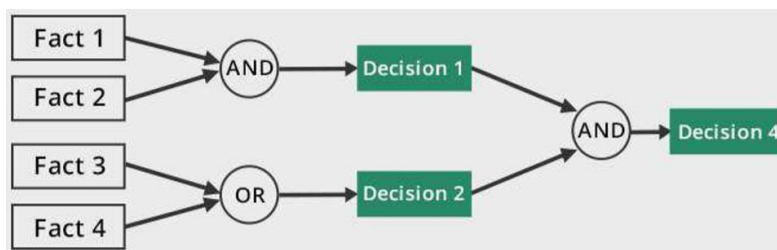
- Forward Chaining
- Backward Chaining

Forward Chaining

It is a strategy of an expert system to answer the question, “What can happen next?”

Here, the inference engine follows the chain of conditions and derivations and finally deduces the outcome. It considers all the facts and rules, and sorts them before concluding to a solution.

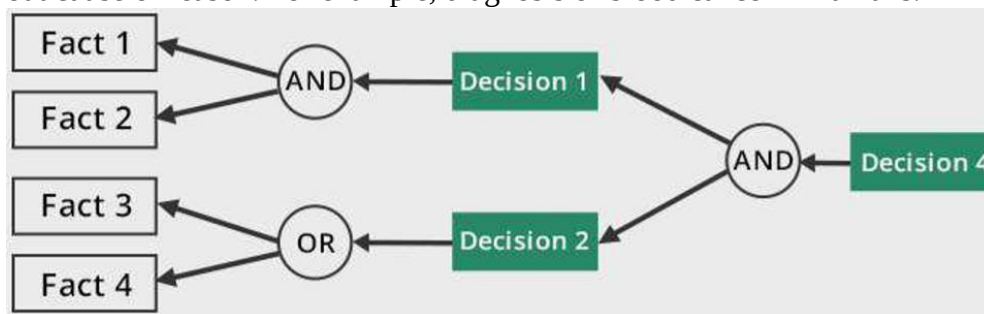
This strategy is followed for working on conclusion, result, or effect. For example, prediction of share market status as an effect of changes in interest rates.



Backward Chaining

With this strategy, an expert system finds out the answer to the question, “Why this happened?”

On the basis of what has already happened, the inference engine tries to find out which conditions could have happened in the past for this result. This strategy is followed for finding out cause or reason. For example, diagnosis of blood cancer in humans.



3. User Interface

User interface provides interaction between user of the ES and the ES itself. It is generally Natural Language Processing so as to be used by the user who is well-versed in the task domain.

The user of the ES need not be necessarily an expert in Artificial Intelligence.

It explains how the ES has arrived at a particular recommendation. The explanation may be in the following forms:

- Natural language displayed on screen
- Verbal narrations in natural language
- Listing of rule numbers displayed on the screen.

The user interface makes it easy to trace the credibility of the deductions.

Requirements of Efficient ES User Interface

- It should help users to accomplish their goals in shortest possible way.

- It should be designed to work for user's existing or desired work practices.
- Its technology should be adaptable to user's requirements; not the other way round.
- It should make efficient use of user input.

Expert Systems Limitations

No technology can offer easy and complete solution. Large systems are costly, require significant development time, and computer resources. ESs have their limitations which include:

- Limitations of the technology
- Difficult knowledge acquisition
- ES are Difficult to maintain
- High Development costs

Applications of Expert System

The following table shows where ES can be applied

Application	Description
Design Domain	Camera lens design, automobile design.
Medical Domain	Diagnosis Systems to deduce cause of disease from observed data, conduction medical operations on humans.
Monitoring Systems	Comparing data continuously with observed system or with prescribed behavior such as leakage monitoring in long petroleum pipeline.
Process Control Systems	Controlling a physical process based on monitoring.
Knowledge Domain	Finding out faults in vehicles, computers.
Finance/Commerce	Detection of possible fraud, suspicious transactions, stock market trading, Airline scheduling, cargo scheduling.

Expert System Technology

There are several levels of ES technologies available. Expert systems technologies include:

1. Expert System Development Environment: The ES development environment includes hardware and tools. They are:
 - Workstations, minicomputers, mainframes
 - High level Symbolic Programming Languages such as LISP and PROLOG.
 - Large databases
2. Tools: They reduce the effort and cost involved in developing an expert system to large extent.
 - Powerful editors and debugging tools with multi-windows.
 - They provide rapid prototyping
 - Have Inbuilt definitions of model, knowledge representation, and inference design.
3. Shells: A shell is nothing but an expert system without knowledge base. A shell provides the developers with knowledge acquisition, inference engine, user interface, and explanation facility. For example, few shells are given below:
 - Java Expert System Shell (JESS) that provides fully developed Java API for creating an expert system.

- Vidwan, a shell developed at the National Centre for Software Technology, Mumbai in 1993. It enables knowledge encoding in the form of IF-THEN rules.

Development of Expert Systems:

The process of ES development is iterative. Steps in developing the ES include:

- Identify Problem Domain
- The problem must be suitable for an expert system to solve it.
- Find the experts in task domain for the ES project.
- Establish cost-effectiveness of the system.

Design the System

- Identify the ES Technology.
- Know and establish the degree of integration with the other systems and databases.
- Realize how the concepts can represent the domain knowledge best.

Develop the Prototype

Form Knowledge Base: The knowledge engineer works to:

- Acquire domain knowledge from the expert.
- Represent it in the form of If-THEN-ELSE rules.

Test and Refine the Prototype

- The knowledge engineer uses sample cases to test the prototype for any deficiencies in performance.
- End users test the prototypes of the ES.

Develop and Complete the ES

- Test and ensure the interaction of the ES with all elements of its environment, including end users, databases, and other information systems.
- Document the ES project well.
- Train the user to use ES.

Maintain the System

- Keep the knowledge base up-to-date by regular review and update.
- Cater for new interfaces with other information systems, as those systems evolve.

Benefits of Expert Systems

- Availability: They are easily available due to mass production of software.
- Less Production Cost: Production cost is reasonable. This makes them affordable.
- Speed: They offer great speed. They reduce the amount of work an individual puts in.
- Less Error Rate: Error rate is low as compared to human errors.
- Reducing Risk: They can work in the environment dangerous to humans.
- Steady response: They work steadily without getting emotional, tensed or fatigued.